

DAY-4

DOCUMENTATION

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1.objective:

To learn the fundamentals of API, AI API and Gemini AI , understand how AI chatbot work, install the Google Generative AI library and create a simple AI chatbot using python.

- What is an API ?

API full form (Application Programming Interface) is a set of rules that allows different software applications to communicate and share data with each other.

1.Why do we use APIs:

- To retrieve data from servers and external systems.
- To connect different applications and enable communication between them.
- To save time and effort.

2.Where are APIs used:

- Weather applications.
- Google maps.
- Payment gateway
- Social media applications.
- AI chatbots.

3.Where do we run API programs:

- Vs code
- Pycharm
- Jupyter Notebook

- Google colab
- IDLE Python

4. Advantages of APIs:

- Easy communication between applications
- Faster development
- Better scalability
- Improved security
- Reduces coding effort

• What Is an AI API ?

AI API full form(Artificial Intelligence Application Programming Interface) is a tool that allows applications to access and use Artificial intelligence services such as answering questions, generating text, translating languages, creating content and more.

1. Where are AI APIs used:

- AI chatbots
- Content generation
- Translation applications
- Customer support
- Education applications

2. Benefits of AI APIs:

- Intelligent responses
- Faster development
- Time saving
- Automation support
- Easy integration

3. Examples:

- Google Gemini API
- OpenAI API
- Anthropic API

● What is an Gemini AI?

Gemini AI is an Artificial Intelligence model developed by ai.google.dev. It can understand and generate human-like text, answer questions, write code, summarize information, and perform many intelligent tasks.

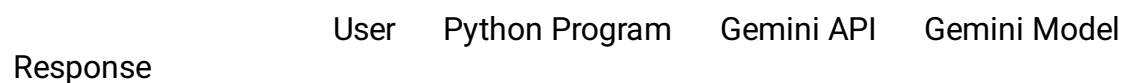
1. Why do we use Gemini AI:

- Answer questions
- Generate ext and content
- Write and explain code
- Summarize information
- Translate languages
- Create AI chatbots

2.Benefits of Gemini AI:

- Provides intelligent answers
- Understands natural language
- Generates content quickly
- Helps automate tasks

3.Working process:



● Difference between normal python program and AI API chatbot

Feature	Normal Python Program	AI API Chatbot
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Logic	Fixed Rules	AI Generated
Responses	Predefined	Dynamic
Flexibility	Limited	Highly flexible
Intelligence	Rule based	AI-Powered
Learning Ability		
Example	No Calculator	Based on training models AI Assistant

- How to install google-generative AI:

Step-1: Open terminal or command prompt.

- Command Prompt
- Terminal
- Vs code Terminal

Step-2: Run the installation command.

- Pip install google-generativeai

Step-3: Wait for installation.

- Python will download and install the required package and dependencies automatically.

Step-4: Verify the installation.

```
import google.generativeai as genai
print("Google Generative AI installed successfully")
```

Conclusion:

APIs enable communication between applications, while AI APIs provide intelligent capabilities. Google's Gemini API allows developers to integrate advanced AI features into their applications. Compared to traditional Python programs, AI chatbots offer dynamic and intelligent responses. The google-generativeai library provides an easy way to access Gemini models and build AI-powered applications.